BIO303 BIOCHEMISTRY II

FINAL TERM PAPER PREPARATION BEST FILE BY

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BS-ZOOLOGY 4TH SEMESTER

SPRING 2022

Enzyme kinetics when substrate concentration is high:

an increase in substrate concentration leads to an increase in the rate of an enzyme-catalyzed reaction. As the enzyme molecules become saturated with substrate, this increase in reaction rate levels off. The rate of an enzyme-catalyzed reaction increases with an increase in the concentration of an enzyme.

Enzyme kinetics when substrate concentration is less:

For an enzyme-catalysed reaction, there is usually a hyperbolic relationship between the rate of reaction and the concentration of substrate, as shown below: (A) At low concentration of substrate, there is a steep increase in the rate of reaction with increasing substrate concentration.

Define Pentose Phosphate Pathway and write the other name of this pathway?

The pentose phosphate pathway takes place in the cytosol of the cell, the same location as glycolysis. The two most important products from this process are the ribose-5-phosphate sugar used to make DNA and RNA, and the NADPH molecules which help with building other molecules.

The hexose monophosphate shunt is the other name of the pentose phosphate pathway.

What are the ribosomal subunits:

Ribosomes contain two different subunits, both of which are required for translation. The small subunit (“40S” in eukaryotes) decodes the genetic message and the large subunit (“60S” in eukaryotes) catalyzes peptide bond formation.

What is 1st order reaction of substrate:

A chemical reaction in which the rate of reaction is directly proportional to the concentration of the reacting substance.

Function of isocitrate lyase in glyoxylate cycle:

Isocitrate lyase (EC 4.1. 3.1), or ICL, is an enzyme in the glyoxylate cycle that catalyzes the cleavage of isocitrate to succinate and glyoxylate. Together with malate synthase, it bypasses the two decarboxylation steps of the tricarboxylic acid cycle (TCA cycle) and is used by bacteria, fungi, and plants.

Disadvantages of triacylglycerols as energy storage:

1= The properties that make triacylglycerols good storage compounds present problems in their role as fuels. 2= Fats must be emulsified before enzymes can digest them. 3= Fats are insoluble in the blood and must be carried in the blood as protein complexes.

Advantages of using triacylglycerols for energy storage:

1= Fats are highly reduced hydrocarbons with a large energy of oxidation. 2= Fats are insoluble molecules that aggregate into droplets. They are unsolvated and no storage mass is water. 3= Fats are chemically inert. They can be stored without fear of unfavorable reactions.

Which ions are formed by the ionization of water and what happened to H+ ion?

Pure Water ls Slightly lonized and Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium reaction $H_2O \leftrightarrow H^+ + OH$

Hydrogen ions formed in water are immediately hydrated to hydronium ions (H_3O^+)

why electric current passing through pure water:

water is capable of conducting electricity due to the dissolved ions and impurities. Pure water carries electrical current as H_3O^+ migrates toward the cathode and OHtoward the anode. In this way electric current passing through pure water

Difference between Oxidation and Reduction

OXIDATION	REDUCTION
Oxidation is a reaction that removes an electron from a substance.	reduction is a reaction that adds electrons to a substance.
Losing electrons	Gaining electrons
Increase in oxidation number	Decrease in oxidation number
For a given compound losing hydrogen	For a given compound gaining hydrogen
This reaction releases energy	This reaction stores energy
oxidizing agents: Ozone, Bleach, peroxide	Common reducing agent is metal

What is Enzyme inhibitor and explain its classes?

Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions. Any substance that can diminish the velocity of an enzyme catalyzed reaction is called an inhibitor. **For example:** aspirin inhibits the enzyme (cyclooxygenase).

Two broad classes of enzyme inhibitors:

1= Reversible 2= Irreversible

1= Reversible inhibitors typically bind to enzymes through noncovalent bonds. 2= In general, irreversible inhibitors, bind to enzymes through covalent bonds.

Others: Competitive inhibitors resemble the substrate and compete for binding to the active site of the enzyme. Noncompetitive inhibitors do not bind at the active site. They bind either free enzyme at a site other than active site or the ES complex.

how malate synthetase through ligh on evolutionary history of glyoxylate cycle.

In recent years, the detection of malate synthase (MS) and isocitrate lyase (ICL), key enzymes involved in the glyoxylate cycle, in some animal tissue has raised questions regarding the evolutionary relationship of enzymes in bacteria and animals and suggests that animals encode alternative enzymes of the cycle that differ in function from known MS and ICL in non-metazoan species.

This bypasses the decarboxylation steps that take place in the TCA cycle, allowing simple carbon compounds to be used in the later synthesis of macromolecules, including glucose. Glyoxylate is subsequently combined with acetyl-CoA to produce malate, catalyzed by malate synthase. Malate is also formed in parallel from succinate by the action of succinate dehydrogenase and fumarase.

Coordinated regulation of glyoxylate and citric acid cycles.

The central molecules between 2 cycles is isocitrate • The translocation of isocitrate between the citric acid cycle and the glyoxylate cycle is controlled by the activity of isocitrate dehydrogenase, which is regulated by reversible phosphorylation. • A specific protein kinase phosphorylates and inactivates the dehydrogenase. This inactivation shunts isocitrate to the glyoxylate cycle, where it begins the synthetic route toward glucose. • A phosphoprotein phosphatase removes the phosphoryl group from isocitrate dehydrogenase, reactivating the enzyme • Reactivation of enzyme sends more isocitrate through the energy-yielding citric acid cycle • The regulatory protein kinase and phosphoprotein phosphatase are separate enzymatic activities of a single polypeptide.

Define pH and write its equation and what is the pH of water at 25°C?

The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the [H+], i. e, the negative logarithm of the [H+]

pH = log 1/[H+]

= -log[H+]

The pH of water at 25°C, in which H+ and OH– ions are present in equal numbers, is 7.0

pH = -log[1 x 10-7]= 7

The symbol p denotes "negative logarithm of" . For each pH unit less than 7.0, the (H+) is increased tenfold; . for each pH unit above 7.0, it is decreased tenfold.

Competitive inhibitors	Non-competitive inhibitors
Its structure is similar to that of substrate.	Its structure is entirely different.
It attaches at the active site of an enzyme.	It attaches at a point other than active site.
The reaction can be reversed by increasing the substrate concentration.	The reaction cannot be reversed by increasing the substrate concentration.

In sharing why phosphate ions are unequal:

Because oxygen is more electronegative than phosphorus, the sharing of electrons is unequal.

Source of fatty acids in cells:

Cells can obtain fatty acid fuels from three sources: 1. Fats consumed in the diet 2. fats stored in cells as lipid droplets 3. fats synthesized in one organ for export to another Some species use all three sources under various circumstances, others use one or two.

Competitive inhibitor and its effect on vmax:

Its structure is similar to that of substrate. It attaches at the active site of an enzyme. The reaction can be reversed by increasing the substrate concentration.

effect on vmax: The effect of a competitive inhibitor is reversed by increasing [S]. At a sufficiently high substrate concentration, the reaction velocity reaches the Vmax as observed in the absence of inhibitor. When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal Vmax. • Therefore, a competitive inhibitor does not decrease Vmax.

Five biological systems:

The reactions going on in biological systems can be categorized into 5 types: 1. Reactions that make or break carbon-carbon bonds 2. Internal rearrangements, isomerizations, and eliminations 3. Free-radical reactions 4. Group transfer 5. Oxidation-reduction reactions.

what is the function of coenzyme a in the pyruvate dehydrogenase complex.

TPP is the coenzyme of pyruvate dehydrogenase, to which it is strictly bound through noncovalent interactions. It is involved in the transfer of hydroxyethyl or “activated aldehyde” groups.

Function of phosphofructokinase:

Phosphofructokinase-1 (PFK-1) catalyzes the transfer of a phosphoryl group from ATP to fructose 6-phosphate to yield fructose 1,6-bisphosphate.

define fermentation and industerial products of fermentation:

fermentation, chemical process by which molecules such as glucose are broken down anaerobically.

Fermentation is a process widely used in yoghurt production, pickles, bread, other bakery and food products, alcoholic biofuels and other beverages like beer, wine, liquors, ethyl alcohol, etc. Other commercial applications of the fermentation process are: Curing of tea. Tanning of leather.

Six names of fermenters:

Simple fermenters (batch and continuous) • Fed batch fermenter • Air-lift or bubble fermenter • Cyclone column fermenter • Tower fermenter • Packed bed bioreactor • photobioreactor.

formula of glucose and atoms number:

The chemical formula of Glucose is C6H12O6. Glucose is a monosaccharide containing an aldehyde group (-CHO). It is made of 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms. And total is 24 atoms.

What are the ligands:

In biochemistry and pharmacology, a ligand (from the Latin ligandum, binding) is a substance (usually a small molecule), • that forms a complex with a biomolecule to serve a biological purpose. In a narrower sense, it is a signal triggering molecule, binding to a site on a target protein • A molecule bound reversibly by a protein is called a ligand. • Ligands include substrates, inhibitors, activators, and neurotransmitters. A ligand may be any kind of molecule, including another protein. • A ligand binds at a site on the protein called the binding site.

chemical transformation in glycolysis:

In glycolysis, glucose is converted into pyruvate. Glucose is a six- memebered ring molecule found in the blood and is usually a result of the breakdown of carbohydrates into sugars. It enters cells through specific transporter proteins that move it from outside the cell into the cell's cytosol.